

Fluid Flow in Pipes

Final Lab Report

Unit Operations Lab 4

November 10th, 2025

Group 2

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1. Executive Summary

The objective of this experiment is to determine the energy loss coefficients, the friction factor (f), the loss coefficient (K), and the flow coefficient (C_f) for various pipe materials and fittings in a fluid flow system by measuring the pressure drop and flow characteristics of water. The data was compared to existing theoretical and empirical correlations. These parameters help predict energy losses and optimize the performance of pumps and pipelines in process systems. Overall, the experiment analyzes the effects of pipe materials and fittings on fluid flow, and it's an integral part of process design.

The experiment was conducted using the Hampden H-6920 Friction Demonstrator to measure pressure losses and friction factors across different pipe materials and fittings.

The results show that the experimental friction factors for smooth pipes aligned with the Darcy-Weisbach and Colebrook correlations, with minor deviations observed at high flow rates. The measured pressure losses at the fittings matched the standard expected values from the fluid data. Overall, the experimental data supported the theoretical models on fluid flow within pipes and fittings. The friction factors of all the materials showed a decreasing trend as the Reynolds number increased. This was in line with the expected transition from laminar to turbulent flow. The orifice flow coefficients remained constant across the tested flow rates, indicating that the measurement of the flow is stable and reliable. The loss of coefficients of fittings, including elbows, bends, expansions, and contractions, was within the standard empirical range, which served to verify that the apparatus was reliable in generating accurate measurements. Any small deviations in the data can be attributed to errors in measurements, surface roughness, and flow instability at high velocities. These findings confirm that the Hampden H-6920 system is not only an appropriate model for explaining fluid transport behavior in industries but also provides

useful information regarding energy losses, facilitating a better understanding of head loss and enabling the design of efficient piping systems.

2. Introduction

This experiment aims to investigate the flow behavior of water in various pipe materials and fittings, considering the impact of inlet flow rate on the energy loss coefficient, friction factor, and flow coefficient. Within chemical engineering processes, it is crucial to consider the flow rate through expansion, contraction, bends, orifice plates, U-bends, and differing materials. Understanding pressure losses in pipes and fittings directly influences equipment sizing, pump selection, and overall energy efficiency of process systems (1).

In most chemical processes, pipe networks are used to transport gases and liquids between individual pieces of equipment, such as reactors, separation units, or storage tanks. Friction loss along pipes, as well as disturbances at fittings, cause energy loss. Accurate prediction of these losses is essential for the design and operation of piping systems. Since analytical models often lack the ability to capture real-world behavior, particularly in high-velocity, turbulent flow, validation through experimental design is necessary to determine the accuracy of theoretical predictions (1).

This experiment utilized a Hampden H-6920 Pipe Friction Demonstrator, which is equipped with a reservoir, pump, rotameter, differential pressure transmitter, and a series of pipes and fittings (1). Specifically, the tests were conducted on copper, stainless steel, and plastic pipes with varying diameters, enabling the evaluation of frictional head loss and local losses. Measurements of differential pressure and temperature at varying flow rates were used to calculate Reynolds numbers and evaluate whether flow conditions were laminar or turbulent.

The experimental design aimed to investigate four key points. Firstly, the calibration of the rotameter was used to verify the conversion factor of the water flow rate. Then, taking four measurements at varying inlet flow rates allows the flow coefficient, C_x , for the orifice meter to be calculated with accuracy. The evaluation of friction factors as a function of Reynolds number in the straight-line pipe section. Finally, the measurement of head loss coefficients (K) for common fittings. The resulting data was compared with theoretical correlations such as the Hagen-Poiseuille relation for laminar flow and the Colebrook equation for turbulent flow, as well as empirical charts like the Moody diagram (1).

This experiment is significant because it connects the principles of momentum transport with practical measurements in fluid mechanics. By analyzing discrepancies between theoretical and experimental results, the study emphasizes the influence of factors such as pipe roughness, measurement uncertainty, and flow instability on energy losses. The findings provide a foundation for understanding real industrial flow systems, where the optimization of energy usage, accurate prediction of pressure drops, and reliable pump selection are essential for cost-effective and efficient operations.

3. Theory

The behavior of fluid flow through pipes and fittings is governed by the principles of conservation of mass and energy, which can be expressed by the continuity and Bernoulli equations. These relationships describe how pressure, velocity, and elevation interact as fluid moves through a system. In pipe systems, energy losses occur due to viscous friction along the walls and turbulence near fittings, resulting in measurable pressure drops. This experiment

applies these theoretical principles to quantify flow resistance in terms of three dimensionless parameters, the friction factor (f), the loss coefficient (K), and the flow coefficient (C_f).

For the orifice meter section, Bernoulli's equation and the continuity equation are combined to relate the pressure drop across the orifice to the volumetric flow rate (Q). This ideal relationship assumes inviscid, steady, and incompressible flow, but real fluids exhibit viscous effects and turbulence. To account for these deviations, a discharge or flow coefficient (C_f) is introduced, yielding the empirical expression for the volumetric flow rate (Eqn 1.1)

The flow coefficient is determined experimentally and typically ranges from 0.6 to 0.9, depending on the orifice geometry and Reynolds number (2). By comparing calculated and tabulated values of C_f , the experiment evaluates whether the orifice behaves consistently under varying flow conditions. For straight pipe sections, the pressure drop resulting from wall friction is described by the following equation (Eqn 1.3)

The friction factor (f) depends on the flow regime and the pipe's relative roughness k_s/D . In laminar flow ($Re < 2100$), f can be derived using $f = 64/Re$. For turbulent flow, both viscous and inertial effects contribute, and f must be obtained from the Colebrook equation of the Moody chart (3). Experimental data for pressure drop, flow rate, and temperature are used to calculate f and compare the results across copper, stainless steel, and plastic pipes of different diameters.

In addition to straight pipes, fittings and elbows, bends, contractions, and expansions produce additional head losses due to localized turbulence and flow separation. These effects are characterized by the head loss due to the pipe fittings equation (Eqn 1.7). Since the loss coefficient primarily depends on the geometry of the fitting rather than the material, comparing experimental and tabulated values of K allows for assessment of system performance. The

equivalent length method can also express the energy loss of a fitting as an equivalent straight-pipe length using (eqn 1.8), enabling an integrated analysis of total system resistance (4)

Together, these relationships form the theoretical foundation for analyzing experimental data. The measured values of pressure drop, temperature, and flow rate were used to calculate Reynolds number, friction factor, and loss coefficients, which can then be compared with theoretical and empirical predictions to evaluate the accuracy of the experimental setup and influence of flow regime, surface roughness, and geometry on fluid resistance.

<u>Table 1: Equations used in Fluid Flow in Pipes Analysis</u>		
<u>Name/Purpose</u>	<u>Equation</u>	<u>Terms Defined</u>
1. Bernoulli's Equation: Describes energy conservation between two points in steady, incompressible, and frictionless flow.	$P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$	P: Pressure (Pa) ρ: Density (kg/m³) v: Velocity (m/s) g: Gravitational acceleration (9.81 m/s²) h: Elevation (m)
2. <u>Orifice Meter</u> : Determines volumetric flow rate through the orifice meter based on measured pressure drop. C_f is the flow	$Q = C_f A_o \sqrt{\frac{2 \Delta P}{\rho}}$	Q: volumetric flowrate (m³/s) C_f: flow coefficient A_o: orifice area (m²) ΔP: pressure drop

coefficient that corrects for real fluid effects.		(Pa) • ρ: fluid density (kg/m³).
3. <u>Mass flow</u> : Relates volumetric flow rate to mass flow rate using fluid density.	$m = \rho Q$	• m: mass flow rate (kg/s) • ρ: density (kg/m³) • Q: volumetric flow rate (m³/s).
4. <u>Darcy-Weisbach equation</u> : Defines pressure drop along a pipe due to wall friction; used to determine the friction factor f .	$\Delta P = f \frac{L}{D} \frac{\rho v^2}{2}$	• ΔP: pressure drop (Pa) • f: Darcy friction factor • L: pipe length (m) • D: pipe diameter (m) • ρ: fluid density (kg/m³) • v: average velocity (m/s).
5. <u>Head loss from pipe friction</u> : Expresses the same energy loss in terms of head (m of fluid) instead of pressure.	$h_L = f \frac{L}{D} \frac{v^2}{2}$	• h_L: head loss (m) • f: friction factor • L: length (m) • D: diameter (m) • v: velocity (m/s)
6. <u>Laminar-pipe friction factor</u> : Defines the friction factor for laminar flow ($Re < 2100$)	$f = \frac{64}{Re}$	• Re: Reynolds number • f: friction factor

7. <u>Reynolds Number:</u> Indicates ratio of inertial forces to viscous forces in a fluid flow	$\Re = \frac{\rho v L}{\mu}$	• μ: dynamic viscosity (Pa·s) • ρ: fluid density (kg/m³) • v: velocity (m/s) • L: Pipe length (m)
8. <u>Head Loss Due to Pipe Fittings:</u> Calculates head loss caused by local disturbances such as elbows, bends, and contractions, where K is the loss coefficient.	$h_L = K \frac{V^2}{2}$	• h_L: head loss from fittings (m) • K: loss coefficient $K = f \frac{L}{D}$ • V: velocity (m/s)

4. Results

A series of experiments was conducted to analyze flow characteristics through different pipe materials and fittings, including plastic, copper, and stainless steel. These were done using pipes of different inner diameters (0.493", 0.622", 1.049" for plastic and stainless steel, and 0.402", 0.666", 1.025" for copper) at flow rates of 10.8, 16.1, and 21.5 LPM. For each trial, the differential pressure across each pipe fitting and pipe section, as well as the fluid temperature, was recorded. These measurements were then used to calculate the flow coefficient (C_f), friction factor (f), and loss coefficient (K) for each configuration. The results are presented in Figures 1-21.

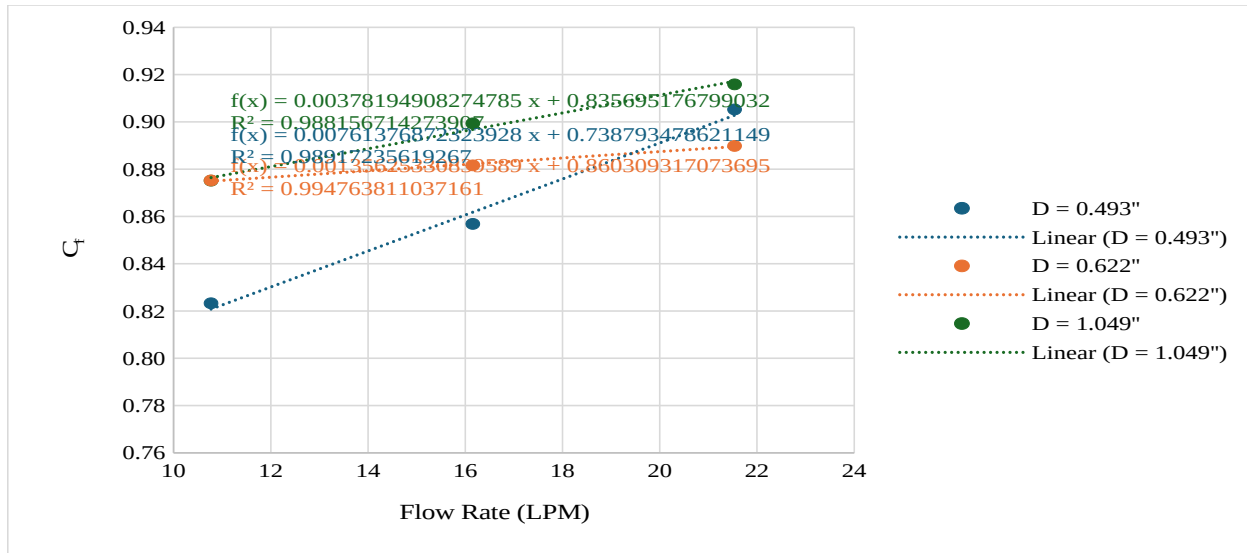
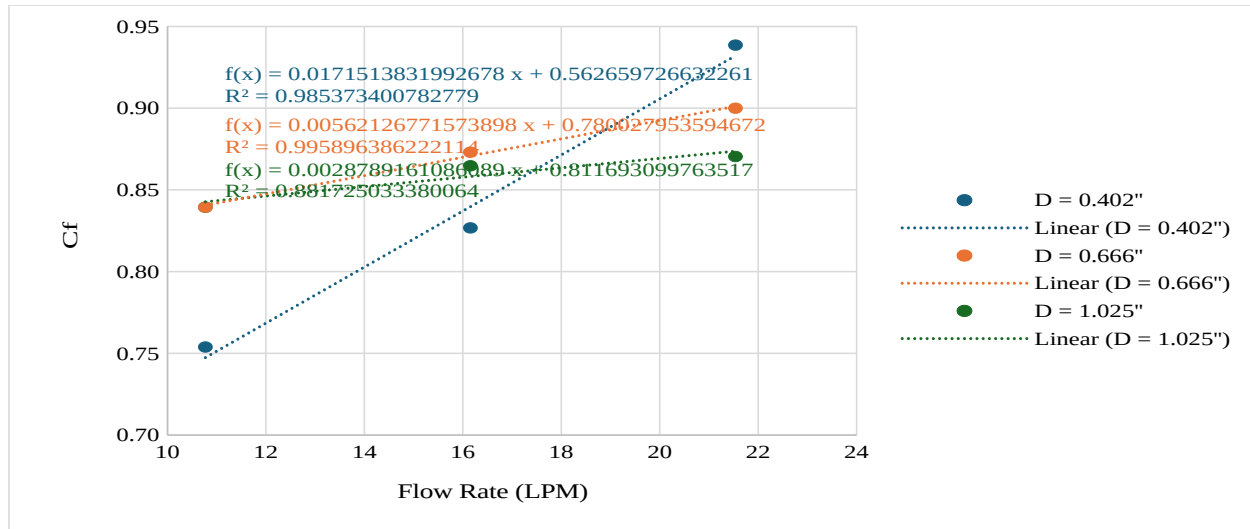


Figure 11: C_f Values at the Orifice Plate for Different Plastic Pipe Diameters

The flow coefficients (C_f) were determined for each pipe using the orifice meter differential pressure drops, as calculated by Equation 1. For plastic pipes, C_f values ranged from 0.82 to 0.92, increasing linearly with the flow rate, as observed in Figure 1. Smaller diameters had lower C_f values due to greater relative resistance, while larger diameters displayed higher coefficients, indicating smoother flow and reduced energy loss.

Figure 22: C_f Values at the Orifice Plate for Different Copper Pipe Diameters as a Function of Flow Rate



As seen in Figure 2, copper pipes followed a similar pattern to plastic pipes, with C_f increasing from 0.75 to 0.94 over the same flow range. Diameters of 0.402", 0.622", and 1.025" each demonstrated strong linear relationships, with larger diameters producing higher coefficients. The slope of the trend lines indicates that C_f increased gradually with flow rate as the fluid became more turbulent.

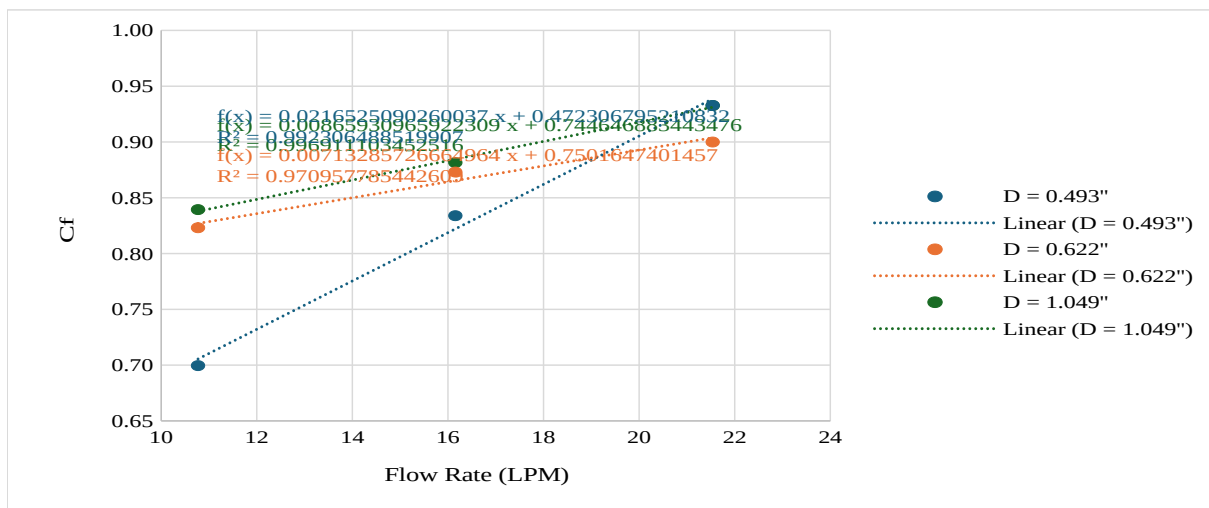


Figure 33: C_f Values at the Orifice Plate for Different Stainless Steel Pipe Diameters as a Function of Flow Rate

Stainless steel pipes produced the highest flow coefficients (Figure 3), ranging from 0.74 to 0.96, with R^2 values between 0.97 and 0.99. The higher C_f values can be attributed to improved flow conditions and reduced velocity irregularities through the stainless-steel pipes.

Following the determination of flow coefficients, Darcy friction factors (f) were calculated from the measured pressure drops at each pipe using Equation 3. They were then plotted as a function of Reynolds number for each material and diameter.

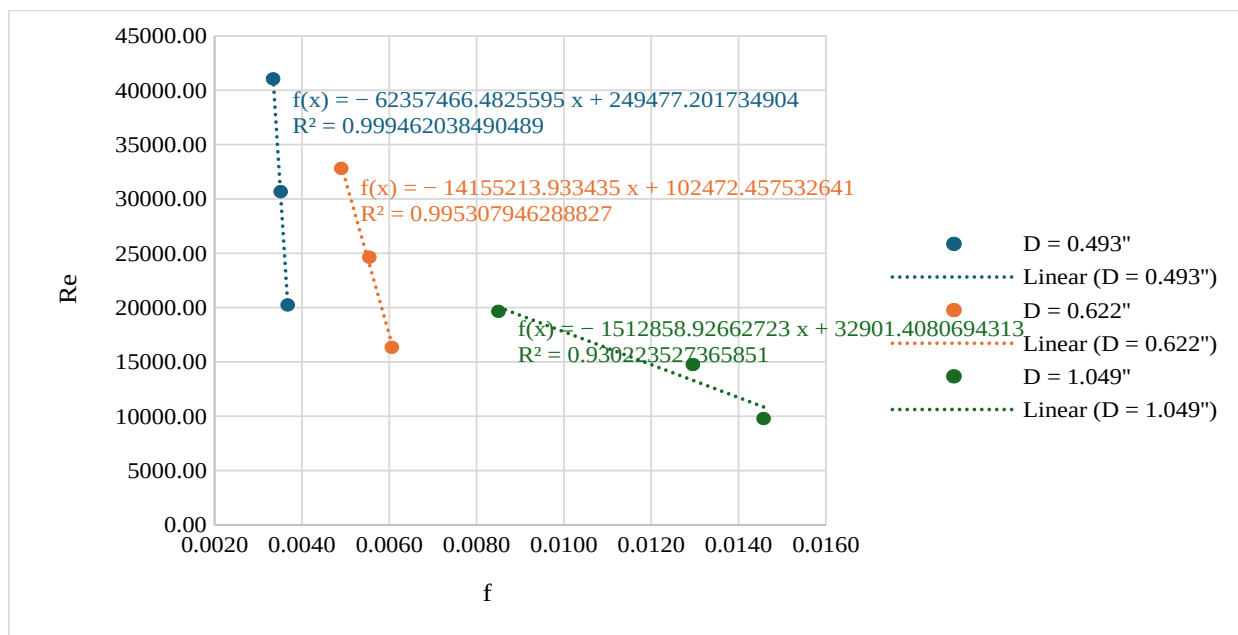


Figure 44: Friction Factors for Plastic Pipes of Different Diameters as a Function of Reynolds Number

As shown in Figure 4, the friction factors for plastic pipes decreased from 0.014 to 0.002 as the Reynolds number increased from 9,800 to approximately 41,000, reflecting the expected turbulent flow behavior. The smallest diameter (0.493") showed the highest friction, while the largest (1.049") exhibited the lowest friction factors.

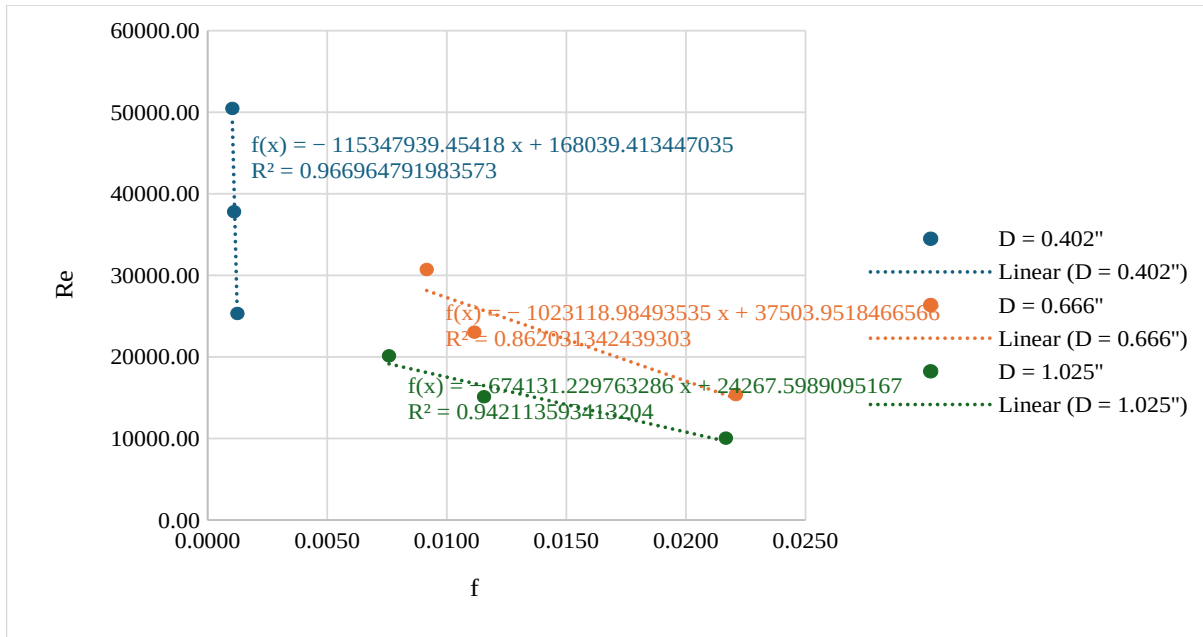


Figure 55: Friction Factors for Copper Pipes of Different Diameters as a Function of Reynolds Number

As seen in Figure 5, copper pipes showed a similar inverse relationship between f and Reynolds number, with friction factors ranging from 0.022 to 0.001. The smaller diameters displayed steeper slopes, suggesting higher sensitivity to flow velocity changes, while the larger diameters produced smoother transitions between data points.

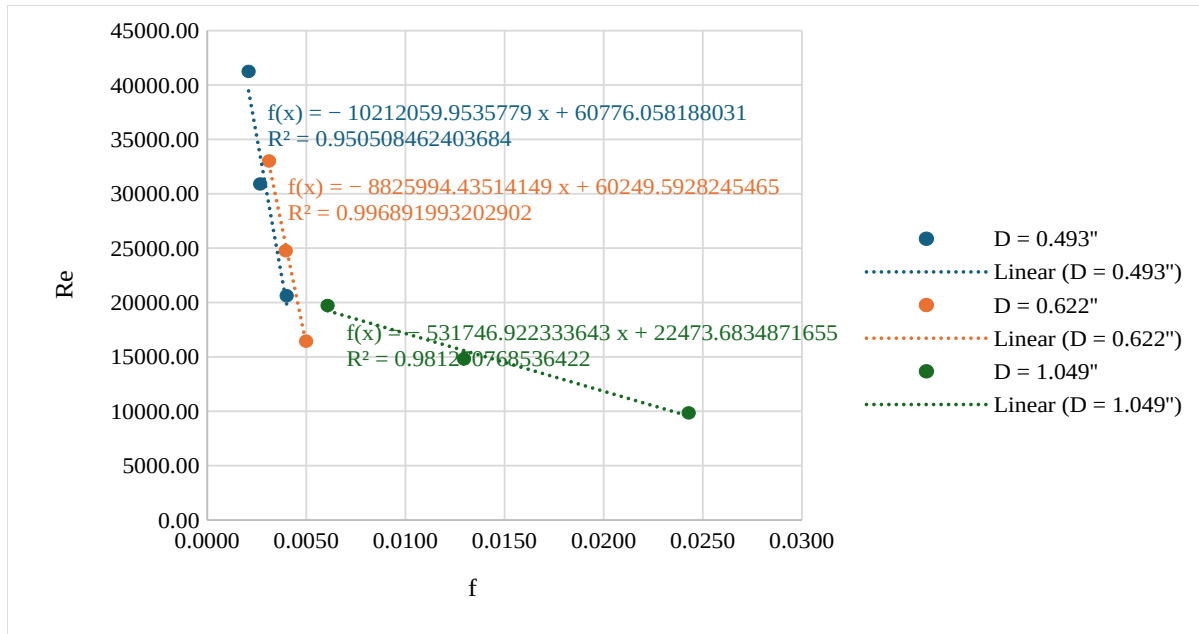


Figure 66: Friction Factors for Stainless Steel Pipes of Different Diameters as a Function of Reynolds Number

Stainless steel pipes had lower friction factors than copper overall, ranging from 0.013 to 0.002, and maintained high correlation coefficients ($R^2 > 0.95$) for all diameters. Stainless steel exhibited the smoothest decline in friction factor as Reynolds number increased, showing the least variability among trials.

To further analyze flow behavior, loss coefficients (K) were determined from the differential pressures for all fittings, including expansions, contractions, bends, and elbows, by using Equation 7.

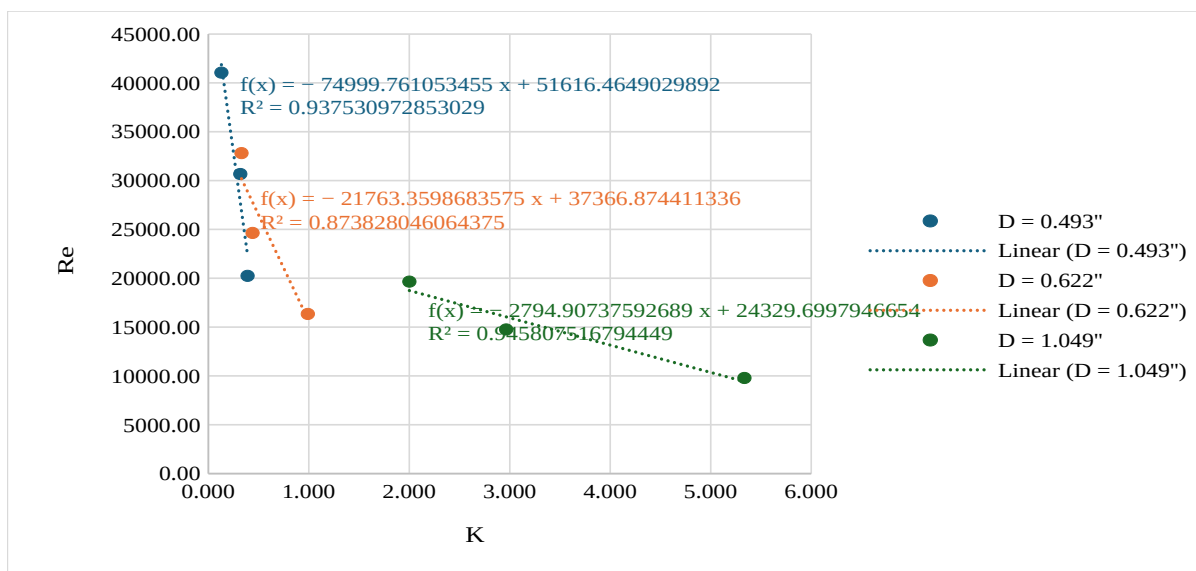


Figure 77: Loss Coefficient (K) for Expansion Pipe as a Function of Reynolds Number while Testing Plastic Pipes of Different Diameters

As shown in Figure 7, for expansion fittings, K values for plastic pipes ranged from 0.13 to 5.33 and decreased with increasing Reynolds number. Copper exhibited a similar trend, with K values from 0.10 to 7.31 (Figure 8), while stainless steel displayed the highest losses overall ranging from 0.11 to 8.00 (Figure 9).

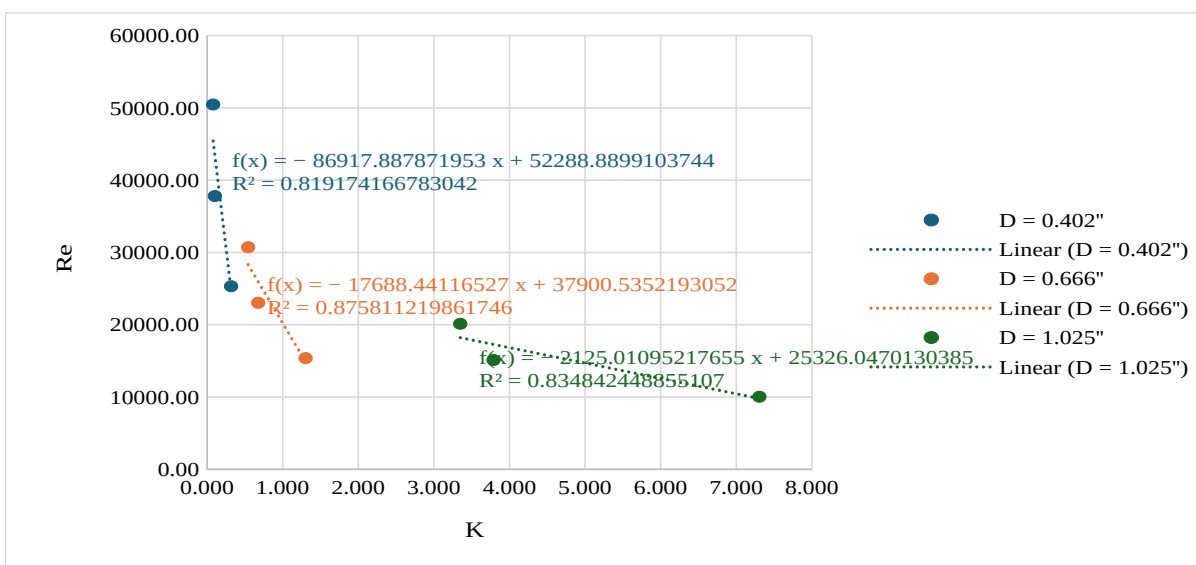


Figure 88: Loss Coefficient (K) for Expansion Pipe as a Function of Reynolds Number while Testing Copper Pipes of Different Diameters

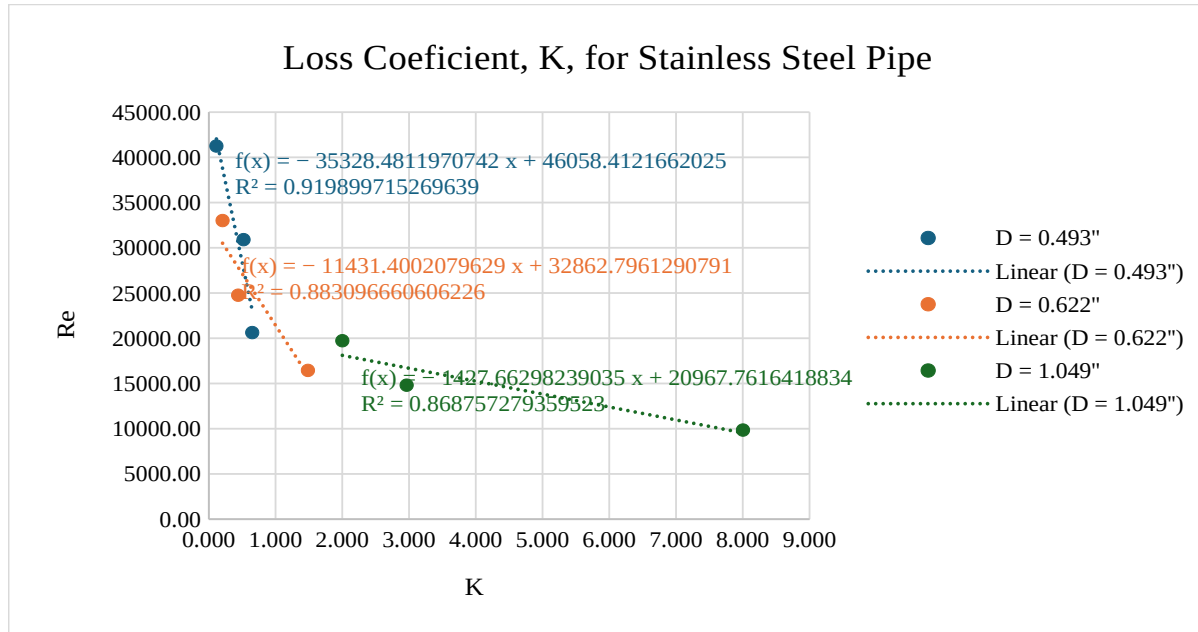


Figure 99: Loss Coefficient (K) for Expansion Fitting as a Function of Reynolds Number while Testing Stainless Steel Pipes of Different Diameters

Contraction fittings also exhibited decreasing K values with increasing Reynolds' number, as shown in Figures 10-12. Plastic ranged from 0.10 to 4.00, copper from 0.04 to 6.10, and stainless steel from 0.08 to 6.67. The results indicate that contraction losses were generally smaller than expansion losses, as flow acceleration tends to streamline the fluid through the narrowing section, thereby reducing turbulence and energy dissipation.

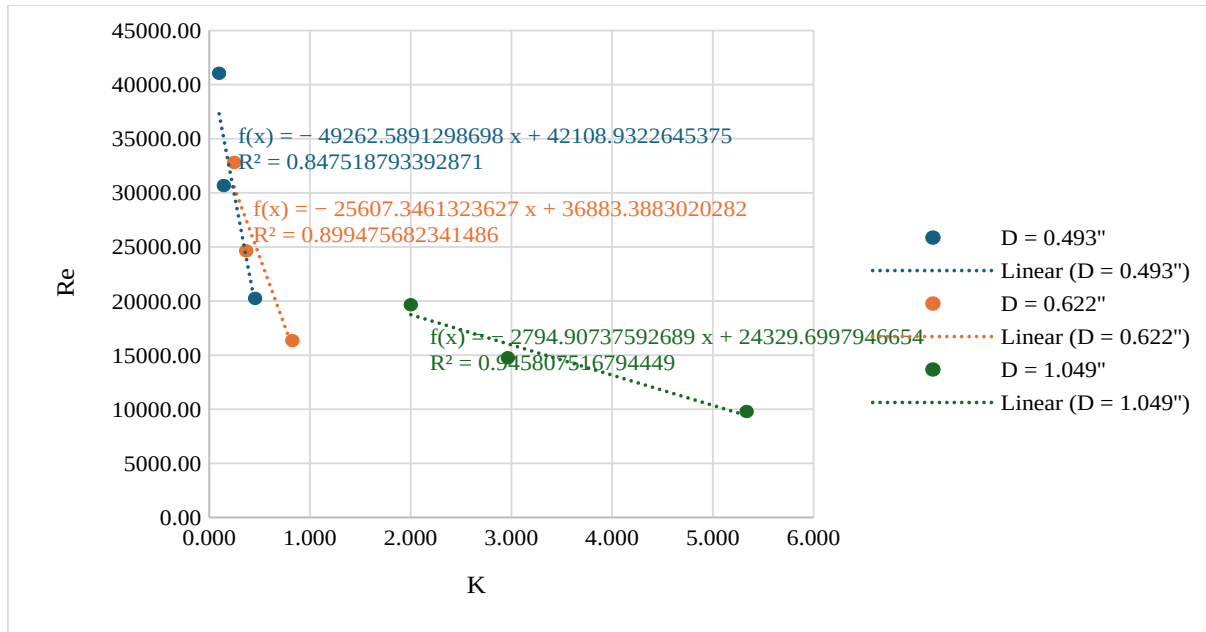


Figure 1010: Loss Coefficient (K) for Contraction Fitting as a Function of Reynolds Number while Testing Plastic Pipes of Different Diameters

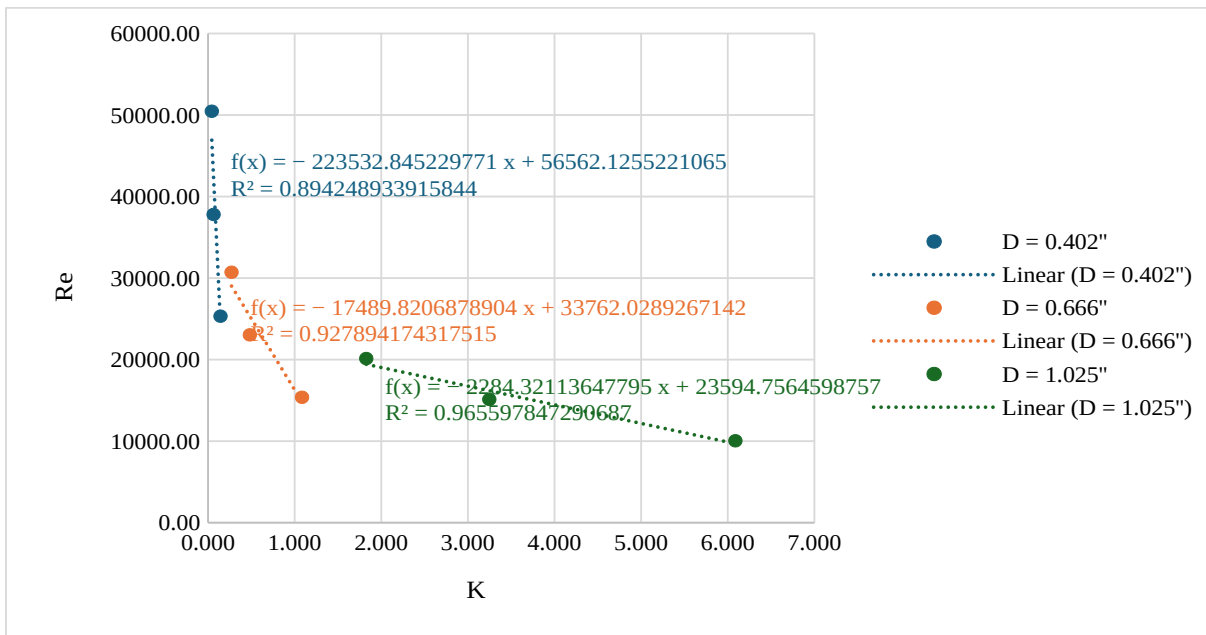


Figure 1111: Loss Coefficient (K) for Contraction Fitting as a Function of Reynolds Number while Testing Copper Pipes of Different Diameters

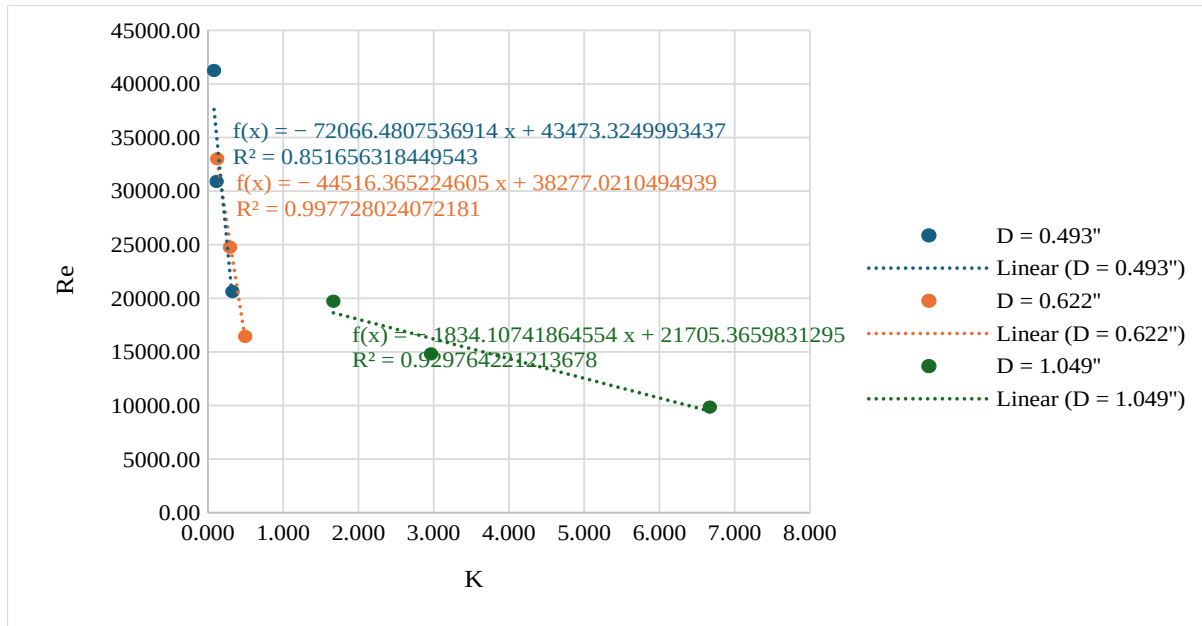


Figure 1212: Loss Coefficient (K) for Contraction Fitting as a Function of Reynolds Number while Testing Stainless Steel Pipes of Different Diameters

The tee fitting displayed higher loss coefficients than the contraction but lower than the expansion. For plastic, copper, and stainless-steel pipes, K values ranged between 0.13-6.67, 0.06-6.10, and 0.09-6.67, respectively, all showing an inverse trend with Reynolds number as seen in Figures 13-15.

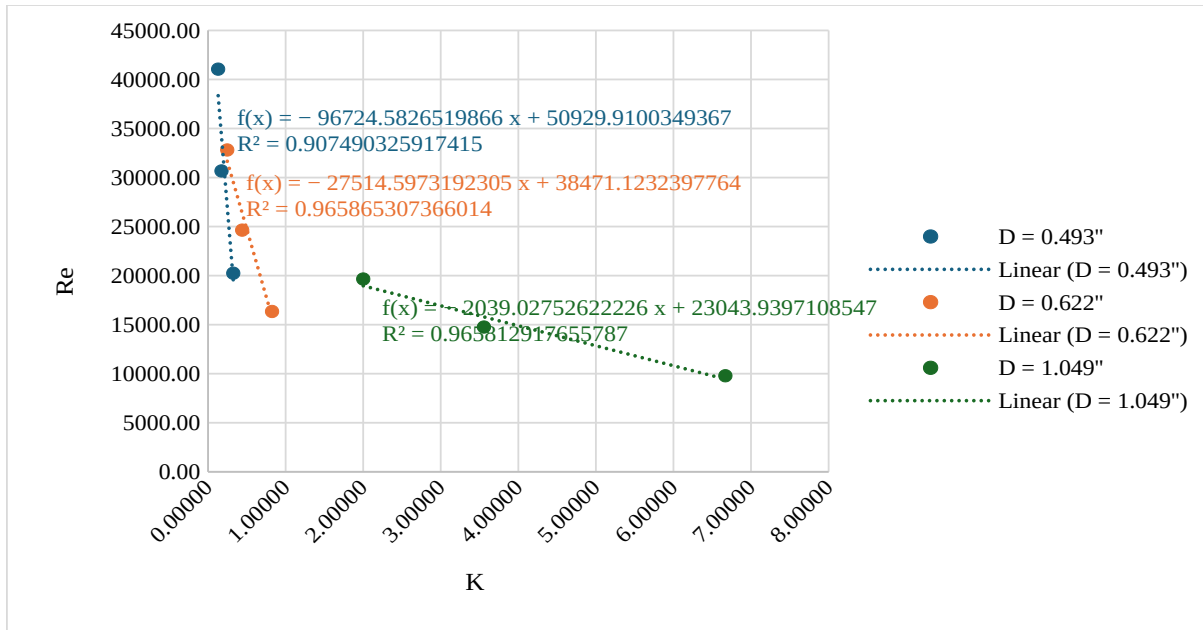


Figure 1313: Loss Coefficient (K) for Tee Fitting as a Function of Reynolds Number while Testing Plastic Pipes of Different Diameters

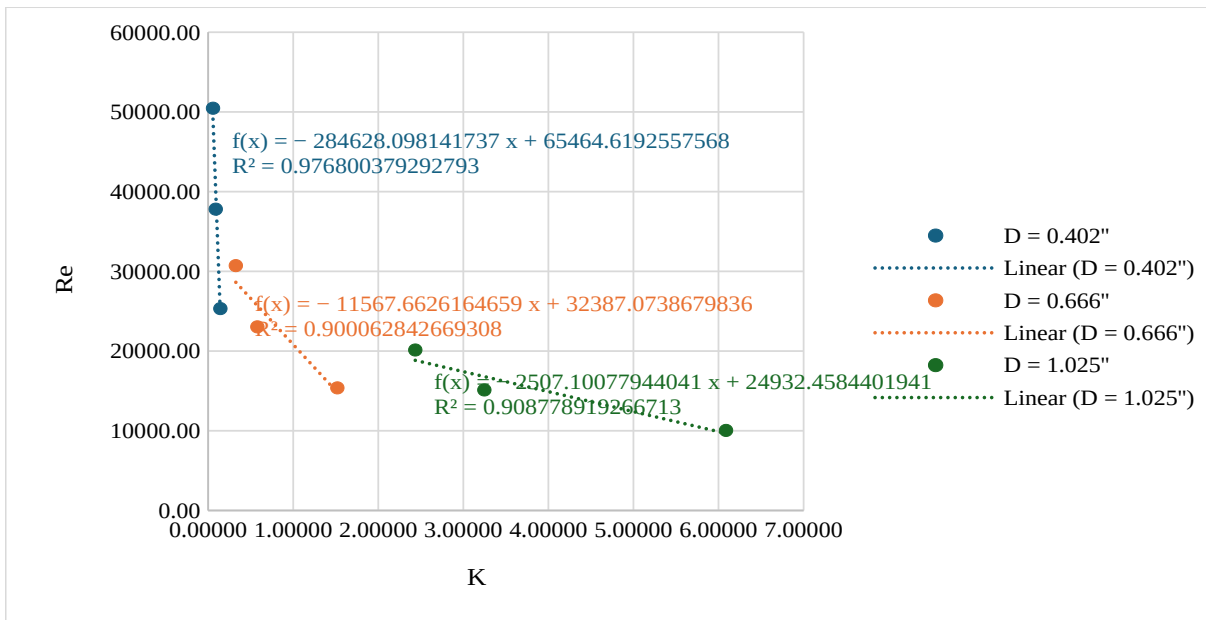


Figure 1414: Loss Coefficient (K) for Tee Fitting as a Function of Reynolds Number while Testing Copper Pipes of Different Diameters

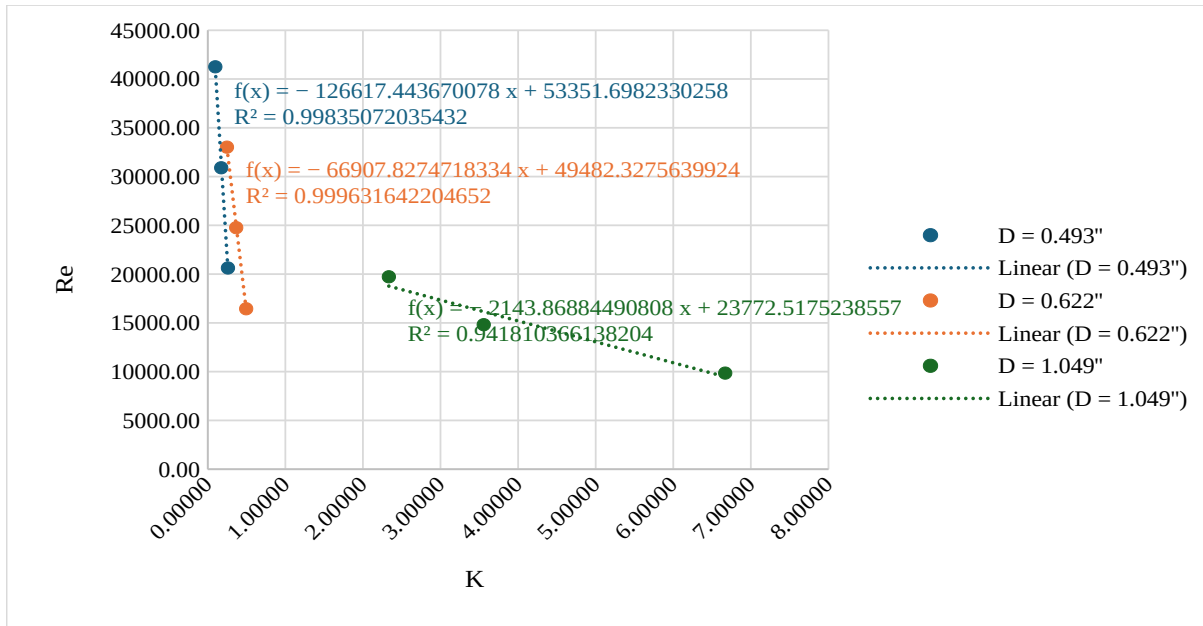


Figure 1515: Loss Coefficient (K) for Tee Fitting as a Function of Reynolds Number while Testing Stainless Steel Pipes of Different Diameters

The 90° Elbow fitting showed a similar pattern, with K values decreasing as flow became more turbulent. Plastic trials ranged from 0.08 to 4.00, copper from 0.03 to 3.65, and stainless steel from 0.05 to 6.67 (Figures 16-18), with higher losses observed in bigger diameters. Overall, the 90° Elbow had the lowest losses out of all the fittings.

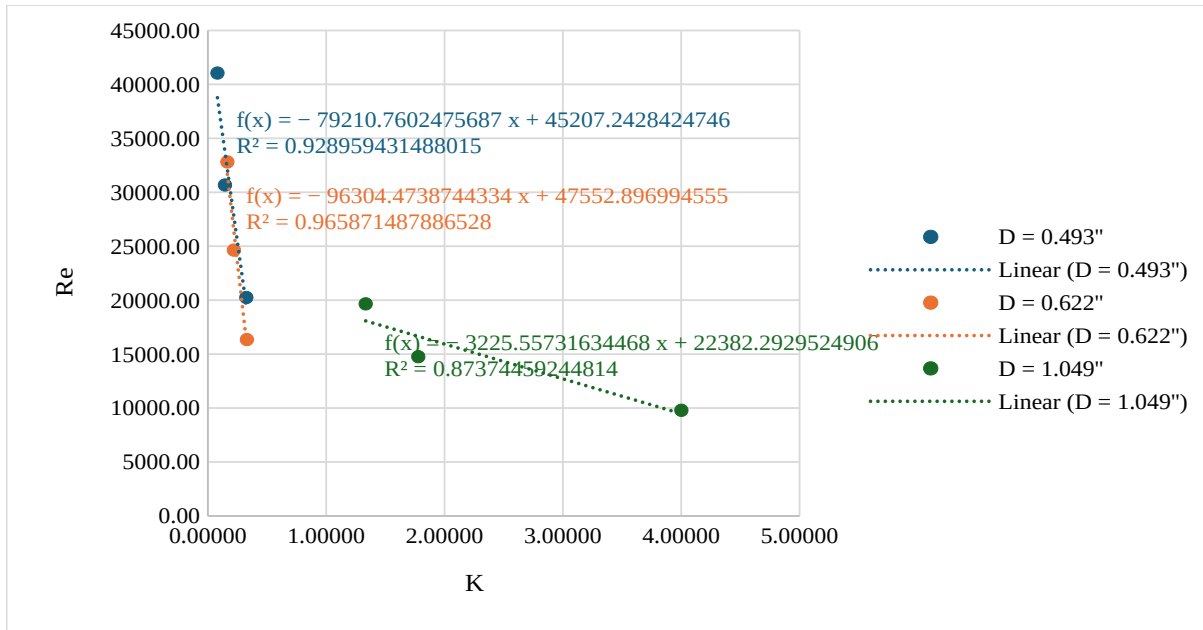


Figure 1616: Loss Coefficient (K) for 90° Elbow as a Function of Reynolds Number while Testing Plastic Pipes of Different Diameters

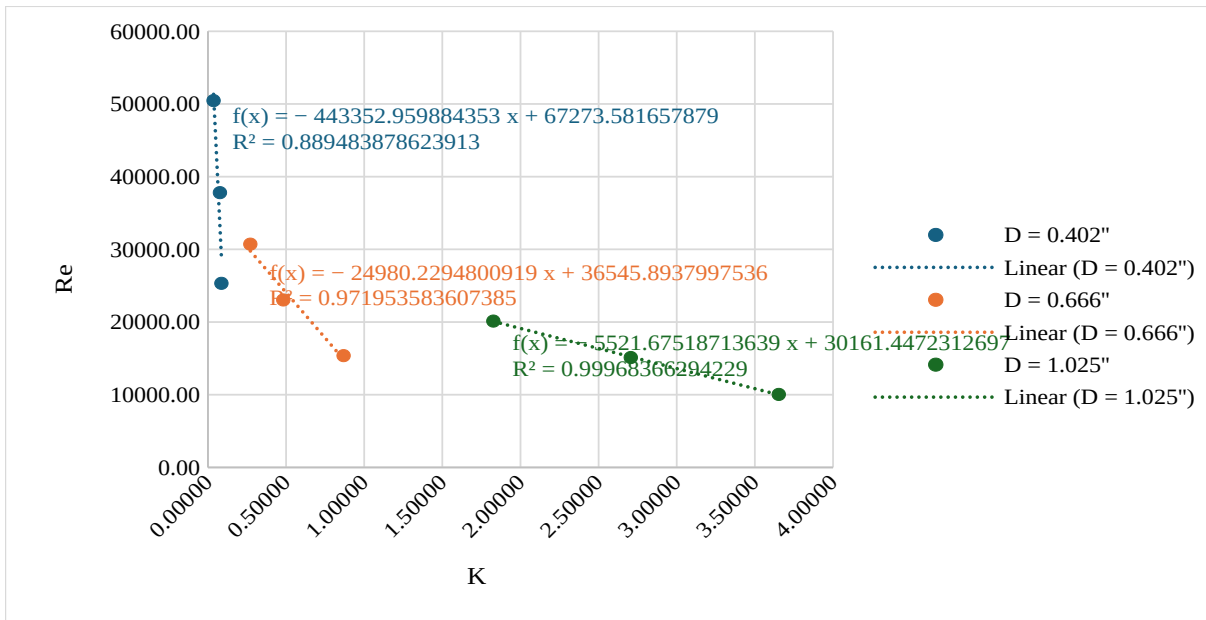


Figure 1717: Loss Coefficient (K) for 90° Elbow as a Function of Reynolds Number while Testing Copper Pipes of Different Diameters

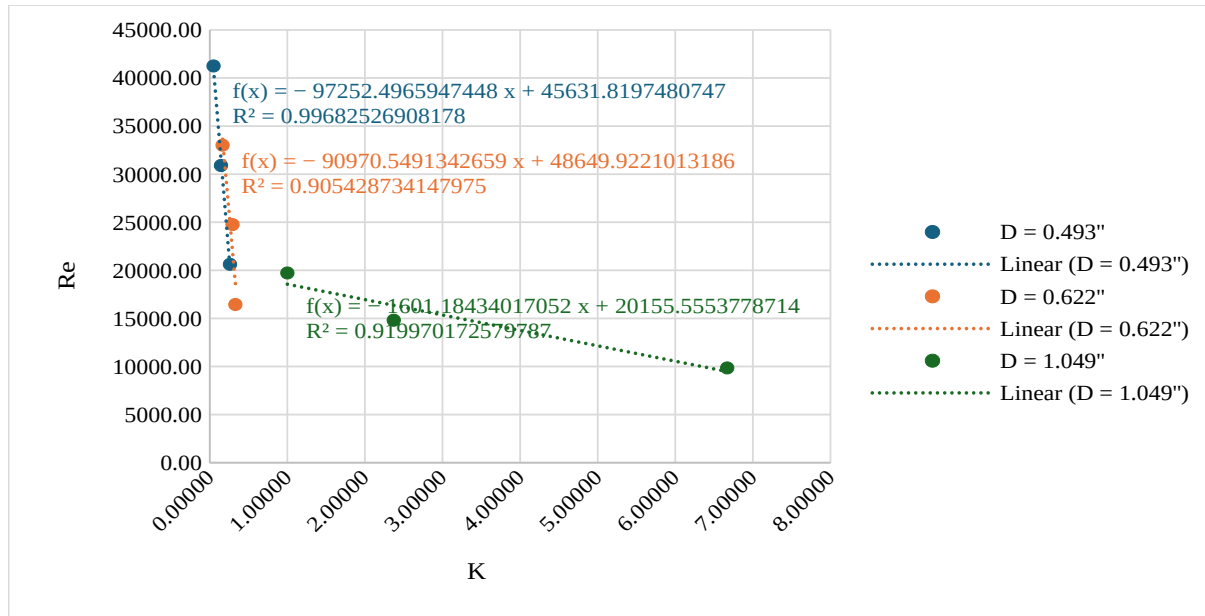


Figure 1818: Loss Coefficient (K) for 90° Elbow as a Function of Reynolds Number while Testing Stainless Steel Pipes of Different Diameters

The 180° bend exhibited higher head losses than the 90° elbow and the tee fittings, with plastic K values ranging from 0.08 to 6.67, copper from 0.03 to 6.09, and stainless steel from 0.08 to 6.67 (Figures 19-21). These values are expected since a full reversal of flow direction produces intense turbulence and a pressure drop.

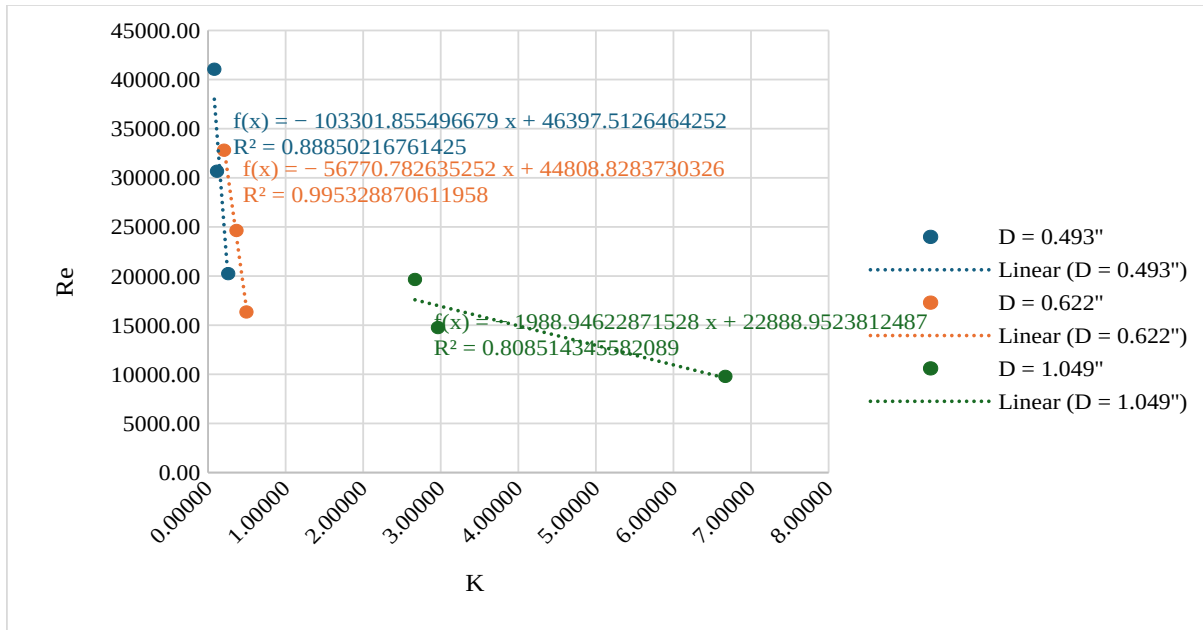


Figure 1919: Loss Coefficient (K) for 180 °Bend as a Function of Reynolds Number while Testing Plastic Pipes of Different Diameters

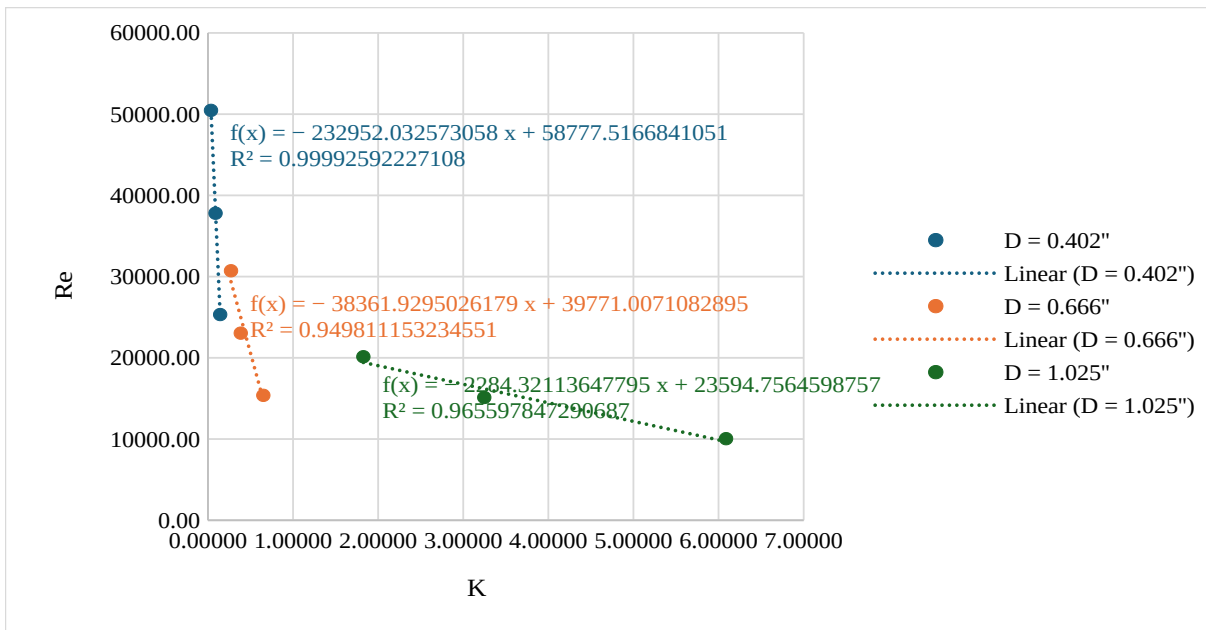


Figure 2020: Loss Coefficient (K) for 180 °Bend as a Function of Reynolds Number while Testing Copper Pipes of Different Diameters

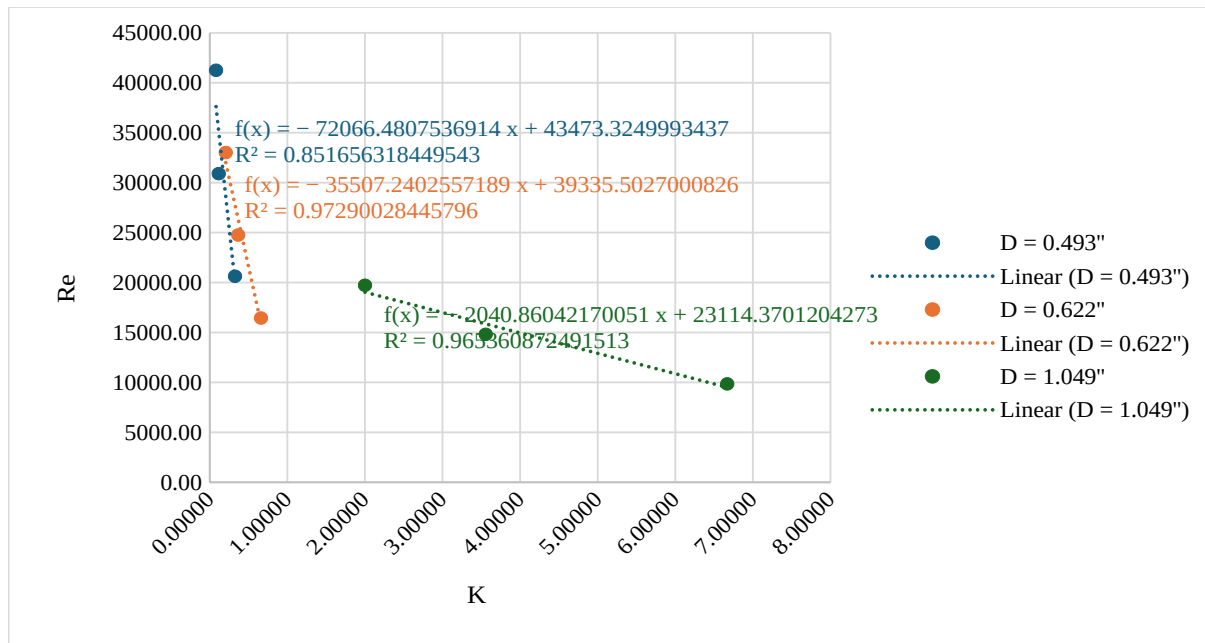


Figure 2121: Loss Coefficient (K) for 180 °Bend as a Function of Reynolds Number while Testing Stainless Steel Pipes of Different Diameters

5. Discussion

The results of this experiment generally align with theoretical expectations and literature values, although some deviations highlight the effect of pipe fittings and materials. The data obtained from the orifice meter calibration, pipe friction analysis, and fittings loss experiment together provided a comprehensive understanding of energy dissipation and flow behavior in pipe systems under various flow conditions.

For the orifice meter, the experimentally determined flow coefficients (C_f) fell within the expected literature range of 0.6-0.9 for sharp-edged orifices, indicating that Bernoulli's equation provided a suitable basis for predicting volumetric flow rate under incompressible, subsonic conditions (1). The data showed a clear trend in which C_f values increased as flow rate increased, which is a relationship that aligns with theoretical expectations that turbulence

enhances the discharge through an orifice by reducing the extent of viscous energy loss near the orifice (1). Additionally, the results demonstrated that the C_f value increased more gradually as pipe diameter increased, consistent with the smaller velocity gradients and reduced contraction effects in larger pipes. Above Reynolds number of approximately 10^4 , the flow coefficients reached a constant, which confirms that viscous effects became negligible and the flow behavior was dominated by inertial forces. The average C_f values obtained from the experiments deviated by less than 5% from the literature mean value of 0.75, validating the calibration of the orifice flowmeter. Minor discrepancies at lower flow rates can be attributed to differential pressure sensitivity limits, small, entrapped air bubbles, and uncertainties in rotameter readings at low Reynolds numbers.

The pipe friction head loss study exhibited clear trends that were consistent with literature correlations. The friction factor (f) decreased as the Reynolds number and flow rate increased, in agreement with theoretical models such as $f = \frac{16}{Re}$ for laminar flow (1) and $f = \frac{0.316}{Re^{0.25}}$ for turbulent flow (1). Furthermore, the results showed that f increased slightly as the pipe diameter decreased, reflecting the effect of increased relative roughness. Copper pipes had higher friction factors compared to stainless steel, which is unexpected given that copper has the least relative roughness of the three materials. This may be due to small-scale experimental factors such as imperfect leveling of the copper pipes, minor air pockets, or inconsistent pressure readings caused by the high thermal conductivity of copper (5). In contrast, plastic pipes exhibited the lowest friction factors, which are likely due to the smoother internal surface and less hydrodynamic disturbance during measurement. Across all pipe materials, the measured

friction factors were within 10% of the values predicted by the Moody chart (1), and the expected slope trend confirmed correct flow characterization.

The fitting loss experiments yielded loss coefficients (K) for a 90° short-radius elbow, an 180 bend, and an expansion-contraction pair that closely matched literature values, ranging between 0.6 and 1.0 (1). The results indicated that loss coefficients for each fitting did not vary significantly between trials or across different pipe materials, which is logical since K depends primarily on the geometry, diameter, and flow rate rather than the material of the pipe. The constant K value with respect to both Reynolds number and material type suggests that the fittings were operating in a fully turbulent regime where local losses are dominated by geometric effects such as flow separation and reattachment, rather than wall friction (1). The measured values were consistent within 15% of tabulated data, with the 90° elbow showing the largest energy loss as expected due to the sharp change in direction. The expansion-contraction fitting had the lowest K , which is typical for symmetric geometries that partially recover pressure downstream (1).

Across all trials, the quantitative agreement with theory was strong; however, several small deviations were noted, which can be attributed to experimental limitations. The average uncertainties were approximately $\pm 3.5\%$ in volumetric flow rate and $\pm 4\%$ in pressure drop, resulting in a propagated uncertainty of $\pm 6\text{--}8\%$ in derived parameters. Systematic errors likely arose from temperature variation, which affects water viscosity and pipe material deviations, rotameter calibration drift, and minor pressure losses in tubing connections. Random errors were caused by transient flow oscillations and a possible lag in the pressure transducer readings. The observation that copper exhibited higher friction factors than expected suggests a local issue within the value.

The theoretical assumptions used in the analysis, such as steady, isothermal, incompressible flow and negligible elevation change, proved sufficiently reliable for this experiment. This is because the deviations from theory were within experimental uncertainty. Overall, the experiment was effective, though greater precision could have been achieved with digital flowmeters and automated pressure logging. Ensuring complete removal of air bubbles and allowing longer stabilization times before recording data would further improve reproducibility.

From these results, several definite conclusions can be drawn. The orifice meter's flow coefficients behave as predicted by classical Bernoulli-based flow models (1) and remain nearly constant at high Reynolds numbers, confirming its suitability for accurate flow measurement. The friction factor trends validate both laminar and turbulent flow correlations, with smooth transitions between regimes, and the relative differences between pipe materials underscore the influence of roughness and diameter on energy dissipation. The fitting losses confirm that K values are governed primarily by geometry, remaining largely independent of material and flow rate once turbulence is established (1). More speculative conclusions concern the anomalously high friction factors in copper pipes, which could arise from unobserved hydrodynamic or thermal effects, and the slight scatter in C_f at low flow rates, potentially due to unsteady pressure conditions.

Overall, the results strongly support classical fluid mechanics theory and demonstrate the practical reliability of empirical correlations for flow measurement and head loss estimation. These findings reinforce the robustness of the Darcy–Weisbach and Bernoulli frameworks in describing real engineering systems. Future investigations could focus on refining pressure measurements with high-frequency digital sensors, extending the study to non-Newtonian fluids, or quantifying the transition region in greater detail to capture the onset of turbulence more

precisely. Despite minor uncertainties, the experiment successfully validated the governing principles of fluid flow in pipes and fittings, confirming both the accuracy of established correlations and the influence of geometry and flow regime on system energy losses.

6. References

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7. Appendix

a. Appendix I: Data Tabulation/Graphs

Table 2: Data for Flow Rotameter Calibration

Time (s)	Volume (mL)	Expected Flow rate (SCFM)	Actual Flow rate (LPM)	Corrected Flow Rate (LPM)
5.81	726	10	7.50	10.77
4.07	925	15	13.64	16.15
5.26	1370	20	15.63	21.54
3.09	1170	25	22.72	26.92
2.51	1250	30	29.88	32.31

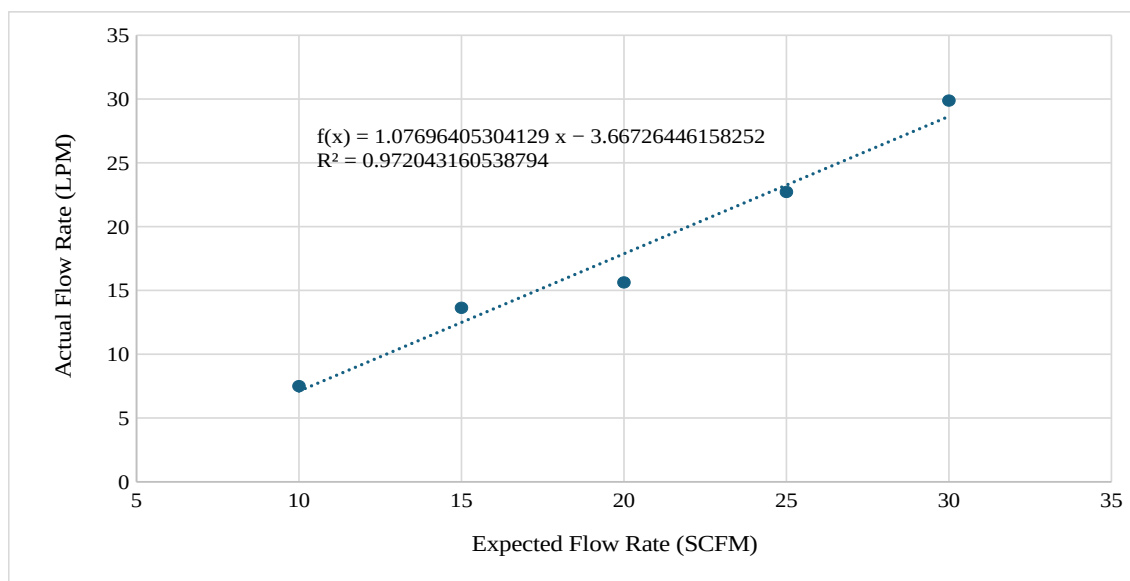
**Figure 22: Rotameter Calibration Curve relating SCFM to LPM**

Table 3: Piping Characteristics

Pipe Material	K _s (mm)	K _s (ft)	Size (in)	Inner Diameter (in)	Length (in)
Copper	0.0015	5E-06	1/2	0.402	72
			3/4	0.666	72
			1 1/8	1.025	72
Plastic	smooth	0	3/8	0.493	72
			1/2	0.622	72
			1	1.049	72
			3/8	0.493	72
			1/2	0.622	72
			1	1.049	72

Table 4: Raw Data for Fluid Flow in Pipes Experiment

Pipe Material	Pipe Size	Trial	Adjusted Flow Rate (LPM)	Expansion Fitting ΔP (psi)	Contraction Fitting ΔP (psi)	Tee Fitting ΔP (psi)	Orifice Meter ΔP (psi)	90° Elbow ΔP (psi)	180° Bend ΔP (psi)	Pipe ΔP (psi)	Temp (°F)
Plastic	3/8"	1	10.77	0.06	0.07	0.05	0.26	0.05	0.04	0.33	76.1
		2	16.15	0.11	0.05	0.06	0.54	0.05	0.04	0.71	76.9
		3	21.54	0.08	0.06	0.08	0.86	0.05	0.05	1.20	77.2
	1/2"	4	10.77	0.06	0.05	0.05	0.23	0.02	0.03	0.17	77.6
		5	16.15	0.06	0.05	0.06	0.51	0.03	0.05	0.35	78.0
		6	21.54	0.08	0.06	0.06	0.89	0.04	0.05	0.55	77.9
	1"	7	10.77	0.04	0.03	0.05	0.23	0.03	0.05	0.03	78.4
		8	16.15	0.05	0.05	0.06	0.49	0.03	0.05	0.06	78.8
		9	21.54	0.06	0.06	0.06	0.84	0.04	0.08	0.07	78.7
Copper	1/2"	10	10.77	0.11	0.05	0.05	0.31	0.03	0.05	0.31	77.7
		11	16.15	0.08	0.05	0.07	0.58	0.06	0.07	0.62	77.3
		12	21.54	0.11	0.06	0.08	0.80	0.05	0.05	1.03	77.4
	3/4"	13	10.77	0.06	0.05	0.07	0.25	0.04	0.03	0.44	78.2
		14	16.15	0.07	0.05	0.06	0.52	0.05	0.04	0.50	78.1
		15	21.54	0.10	0.05	0.06	0.87	0.05	0.05	0.73	78.1
	1-1/8"	16	10.77	0.06	0.05	0.05	0.25	0.03	0.05	0.05	78.6
		17	16.15	0.07	0.06	0.06	0.53	0.05	0.06	0.06	78.9
		18	21.54	0.11	0.06	0.08	0.93	0.06	0.06	0.07	78.8

Stainless Steel	3/8"	19	10.77	0.10	0.05	0.04	0.36	0.04	0.05	0.36	77.6
		20	16.15	0.18	0.04	0.06	0.57	0.05	0.04	0.54	77.5
		21	21.54	0.07	0.05	0.06	0.81	0.03	0.05	0.75	77.6
	1/2"	22	10.77	0.09	0.03	0.03	0.26	0.02	0.04	0.14	78.1
		23	16.15	0.06	0.04	0.05	0.52	0.04	0.05	0.25	78.4
		24	21.54	0.05	0.03	0.06	0.87	0.04	0.05	0.35	78.4
	1"	25	10.77	0.06	0.05	0.05	0.25	0.05	0.05	0.05	78.9
		26	16.15	0.05	0.05	0.06	0.51	0.04	0.06	0.06	79.1
		27	21.54	0.06	0.05	0.07	0.81	0.03	0.06	0.05	79.0

Table 5: Flow Coefficients, Friction Factors, And Reynolds Numbers for Each Experiment Trial

Pipe Material	Trial	Adjusted Flow Rate (m ³ /s)	V (m/s)	Water Density (g/L)	Water Viscosity, μ (Pa/m ²)	C_f	f	Re
Plastic	1	0.00018	1.46	997.30	0.00090	0.82	0.0037	20245.3
	2	0.00027	2.19	997.19	0.00089	0.86	0.0035	30671.7
	3	0.00036	2.92	997.15	0.00089	0.91	0.0033	41050.7
	4	0.00018	0.92	997.09	0.00088	0.88	0.0061	16344.5
	5	0.00027	1.37	997.03	0.00088	0.88	0.0055	24638.1
	6	0.00036	1.83	997.04	0.00088	0.89	0.0049	32810.1
	7	0.00018	0.32	996.90	0.00087	0.88	0.0146	9789.6
	8	0.00027	0.48	996.91	0.00087	0.90	0.0130	14757.2
	9	0.00036	0.64	996.91	0.00087	0.92	0.0085	19653.6
Copper	10	0.00018	2.19	997.07	0.00088	0.75	0.0012	25324.2
	11	0.00027	3.29	997.13	0.00089	0.83	0.0011	37799.9
	12	0.00036	4.38	997.12	0.00088	0.94	0.0010	50462.0
	13	0.00018	0.80	997.00	0.00088	0.84	0.0221	15376.2
	14	0.00027	1.20	997.01	0.00088	0.87	0.0112	23035.6
	15	0.00036	1.60	997.01	0.00088	0.90	0.0092	30714.2
	16	0.00018	0.34	996.94	0.00087	0.84	0.0217	10040.9
	17	0.00027	0.51	996.90	0.00087	0.86	0.0116	15116.2
	18	0.00036	0.67	996.91	0.00087	0.87	0.0076	20129.6
Stainless Steel	19	0.00018	1.46	997.09	0.00088	0.70	0.0040	20626.5
	20	0.00027	2.19	997.10	0.00088	0.83	0.0027	30901.5
	21	0.00036	2.92	997.09	0.00088	0.93	0.0021	41252.9
	22	0.00018	0.92	997.01	0.00088	0.82	0.0050	16445.7
	23	0.00027	1.37	996.97	0.00087	0.87	0.0040	24760.7
	24	0.00036	1.83	996.97	0.00087	0.90	0.0031	33014.3
	25	0.00018	0.32	996.90	0.00087	0.84	0.0243	9850.5

	26	0.00027	0.48	996.87	0.00087	0.88	0.0130	14812.8
	27	0.00036	0.64	996.88	0.00087	0.93	0.0061	19725.6

<u>Table 6: Head Loss Coefficients for Each Type of Fitting</u>						
Pipe Material	Trial	K (Expansion)	K (Contraction)	K (Tee)	K (90° Elbow)	K (180° Bend)
Plastic	1	0.390	0.455	0.325	0.325	0.260
	2	0.318	0.145	0.173	0.145	0.116
	3	0.130	0.098	0.130	0.081	0.081
	4	0.990	0.825	0.825	0.330	0.495
	5	0.440	0.367	0.440	0.220	0.367
	6	0.330	0.248	0.248	0.165	0.206
	7	5.335	4.002	6.669	4.002	6.669
	8	2.964	2.964	3.557	1.778	2.964
	9	2.001	2.001	2.001	1.334	2.668
Copper	10	0.317	0.144	0.144	0.086	0.144
	11	0.102	0.064	0.090	0.077	0.090
	12	0.079	0.043	0.058	0.036	0.036
	13	1.302	1.085	1.519	0.868	0.651
	14	0.675	0.482	0.579	0.482	0.386
	15	0.543	0.271	0.326	0.271	0.271
	16	7.306	6.088	6.088	3.653	6.088
	17	3.788	3.247	3.247	2.706	3.247
	18	3.349	1.827	2.435	1.827	1.827
Stainless Steel	19	0.651	0.325	0.260	0.260	0.325
	20	0.520	0.116	0.173	0.145	0.116
	21	0.114	0.081	0.098	0.049	0.081
	22	1.485	0.495	0.495	0.330	0.660
	23	0.440	0.293	0.367	0.293	0.367
	24	0.206	0.124	0.248	0.165	0.206
	25	8.003	6.669	6.669	6.669	6.669
	26	2.964	2.964	3.557	2.371	3.557
	27	2.001	1.667	2.334	1.000	2.001

Table 7: Equivalent Lengths of Diameter for Each Type of Fitting

Pipe Material	Trial	Expansion L _e (m)	Contraction L _e (m)	Tee L _e (m)	90° Elbow L _e (m)	180° Bend L _e (m)
Plastic	1	1.33	1.55	1.11	1.11	0.89
	2	1.13	0.52	0.62	0.52	0.41
	3	0.49	0.37	0.49	0.30	0.30
	4	2.58	2.15	2.15	0.86	1.29
	5	1.25	1.05	1.25	0.63	1.05
	6	1.06	0.80	0.80	0.53	0.67
	7	9.75	7.32	12.19	7.32	12.19
	8	6.10	6.10	7.32	3.66	6.10
	9	6.27	6.27	6.27	4.18	8.36
Copper	10	2.60	1.18	1.18	0.71	1.18
	11	0.94	0.59	0.83	0.71	0.83
	12	0.78	0.43	0.57	0.36	0.36
	13	1.00	0.83	1.16	0.67	0.50
	14	1.02	0.73	0.88	0.73	0.59
	15	1.00	0.50	0.60	0.50	0.50
	16	8.78	7.32	7.32	4.39	7.32
	17	8.53	7.32	7.32	6.10	7.32
	18	11.50	6.27	8.36	6.27	6.27
Stainless Steel	19	2.03	1.02	0.81	0.81	1.02
	20	2.44	0.54	0.81	0.68	0.54
	21	0.68	0.49	0.59	0.29	0.49
	22	4.70	1.57	1.57	1.05	2.09
	23	1.76	1.17	1.46	1.17	1.46
	24	1.05	0.63	1.25	0.84	1.05
	25	8.78	7.32	7.32	7.32	7.32
	26	6.10	6.10	7.32	4.88	7.32
	27	8.78	7.32	10.24	4.39	8.78

b. Appendix II: Error Analysis

This experiment was subject to both random and systematic errors that influenced the accuracy and precision of the calculated friction factors, flow coefficients, and loss coefficients. The precision of the results was primarily affected by random measurement variations in pressure, flowrate, and temperature. Small fluctuations in speed and minor instability in flow control valves may have caused natural scatter in the differential pressure readings at each flowrate. Similarly, small changes in the water temperature may alter the viscosity and density, introducing additional variability in the Reynolds number and consequently the calculated values of f , K , and C_f . Corrosion of the pipes can also play a major role in the introduction of error on the results. As seen with the copper pipes, potential corrosion may have artificially increased the friction factor, which deviates from theoretical expectations since copper has a low relative roughness. In short, degradation of the pipe materials is another thing that may impact results.

Random error can be quantified using the standard deviation, which measures the degree of spread among repeated readings and is calculated as follows.

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

Differences in the reproducibility of the pressure drop and flowrate readings are reflected by the observed variation in calculated coefficients across trials as shown in the table below.

Table 8 2: Standard Deviations for Calculated Data						
f	C_f	K (Expansion)	K (Contraction)	K (Tee)	K (90° Elbow)	K (180° Bend)
0.0002	0.0412	0.0336	0.0486	0.0256	0.0316	0.0237
0.0006	0.0073	0.0884	0.0762	0.0735	0.0210	0.0362

0.0031	0.0205	0.4290	0.2502	0.5943	0.3573	0.5574
0.0001	0.0930	0.0327	0.0133	0.0109	0.0067	0.0135
0.0070	0.0303	0.1014	0.1056	0.1572	0.0757	0.0487
0.0073	0.0165	0.5422	0.5425	0.4795	0.2284	0.5425
0.0010	0.1170	0.0700	0.0330	0.0203	0.0265	0.0330
0.0009	0.0390	0.1703	0.0465	0.0309	0.0217	0.0575
0.0092	0.0467	0.8059	0.6490	0.5588	0.7394	0.5943

c. Appendix III: Sample Calculations

1) Orifice flow coefficient C_f of copper in trial 10

$$C_f = \frac{Q}{A_o \sqrt{2 \frac{\Delta P}{\rho}}}$$

$$A_o = \frac{\pi d_o^2}{4} = \frac{\pi \times (0.0121)^2}{4} = 0.00011499 \text{ m}^2$$

$$\sqrt{2} \frac{\Delta P}{\rho} = \sqrt{2} \cdot \frac{2137.38}{997.07} = 2.07$$

$$C_{f,c10} = \frac{0.00018}{(0.00011499) \times (2.07)} = 0.75$$

2) Pipe velocity V and Reynolds number Re of copper in trial 10

$$A = \frac{\pi d_o^2}{4} = \frac{\pi \times (0.01021)^2}{4} = 0.0000819 \text{ m}^2$$

$$V = \frac{Q}{A} = \frac{0.00018}{0.0000819} = 2.19 \frac{\text{m}}{\text{s}}$$

- o Reynolds number:

$$Re_{c10} = \frac{\rho v L}{\mu} = \frac{997.07 \times 2.19 \times 0.01021}{0.0008813} = 25324.23$$

- 3) Darcy friction factor f for Trial 10:

$$f = \frac{\Delta P D}{2 L \rho v^2}$$

$$f_{c10} = \frac{2137.38 \times 0.01021}{2 \times 1.8288 \times 997.07 \times (2.19)^2} = 0.0012$$

- o Pressure loss over the section:

$$h_{L,10} = \frac{2137.38}{997.07} = 0.00215 \text{ Pa}$$

- 4) Fitting Loss Coefficient K for Trial 10 of Expansion Fitting:

$$K = \frac{2 * \Delta P}{\rho v^2}$$

$$K = \frac{2 \times 758.42}{997.07 \times 2.19^2} = 0.317$$

- o Pressure loss for the fitting:

$$h_{L,10} = \frac{758.2}{997.07} = 0.788 \text{ Pa}$$